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(54) **METHOD AND APPARATUS FOR
IMPROVEMENT OF VIDEO CODING USING
MERGE WITH MVD MODE WITH
TEMPLATE MATCHING**

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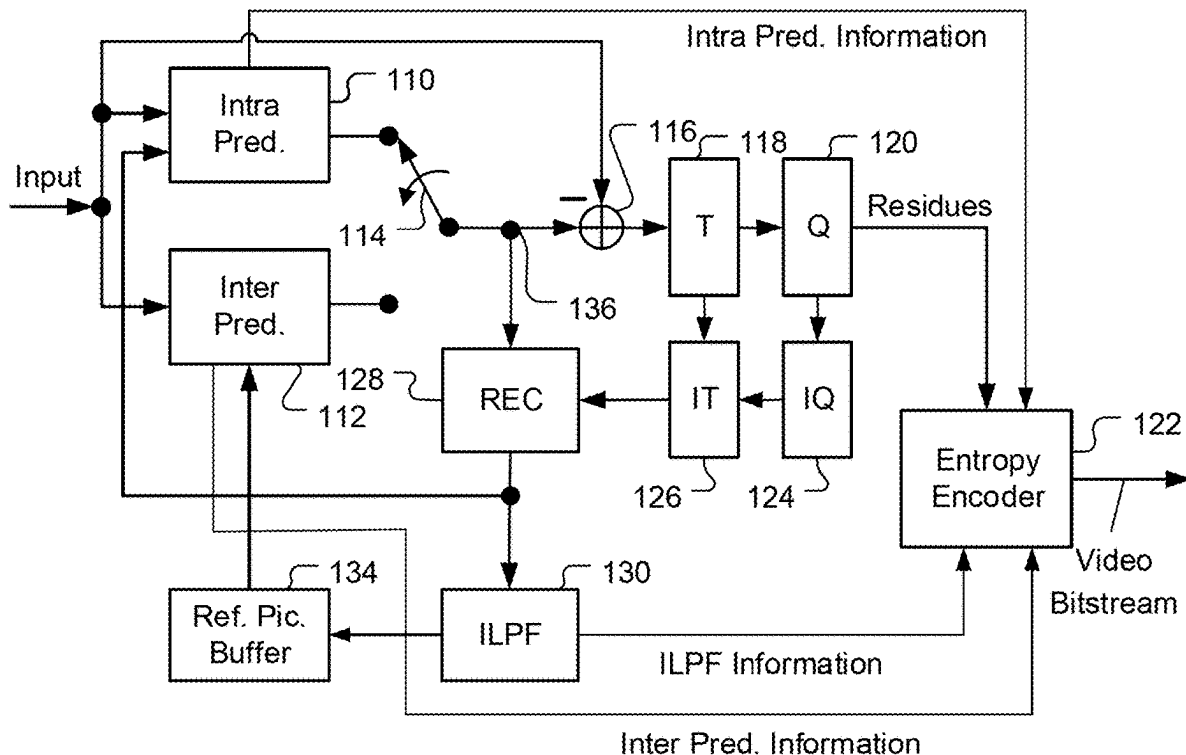
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(57)

ABSTRACT

A method and apparatus for video coding using MMVD mode are disclosed. According to this method, a first expanded merge MV is determined for the current block, where the first expanded merge MV is derived by adding a first selected offset from a first set of offsets to a base MV, and whether the first expanded merge MV is applied to a first reference picture in L0 or a second reference picture in L1 is determined implicitly by the decoder side, or the first expanded merge MV is applied to the first reference picture in the L0 and a second expanded merge MV is applied to the second reference picture in the L1. The current block is encoded or decoded by using motion information comprising the first expanded merge MV. According to another method, separate MVDs are used for reference pictures in different reference lists.



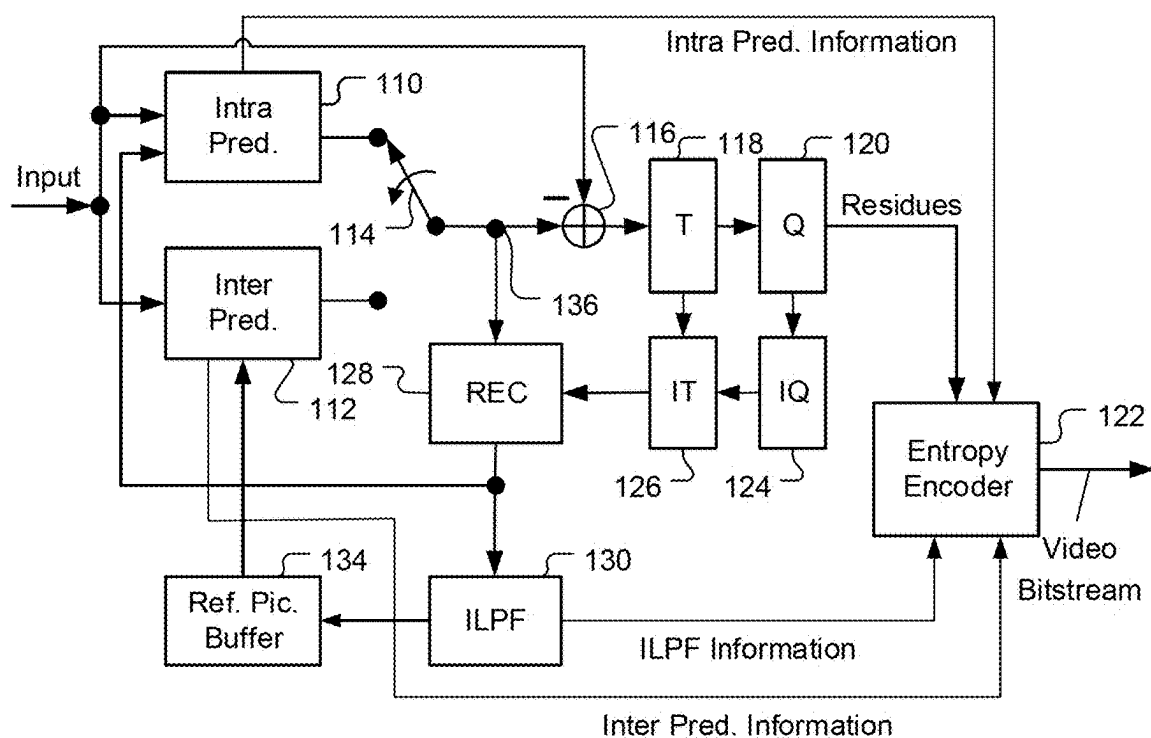


Fig. 1A

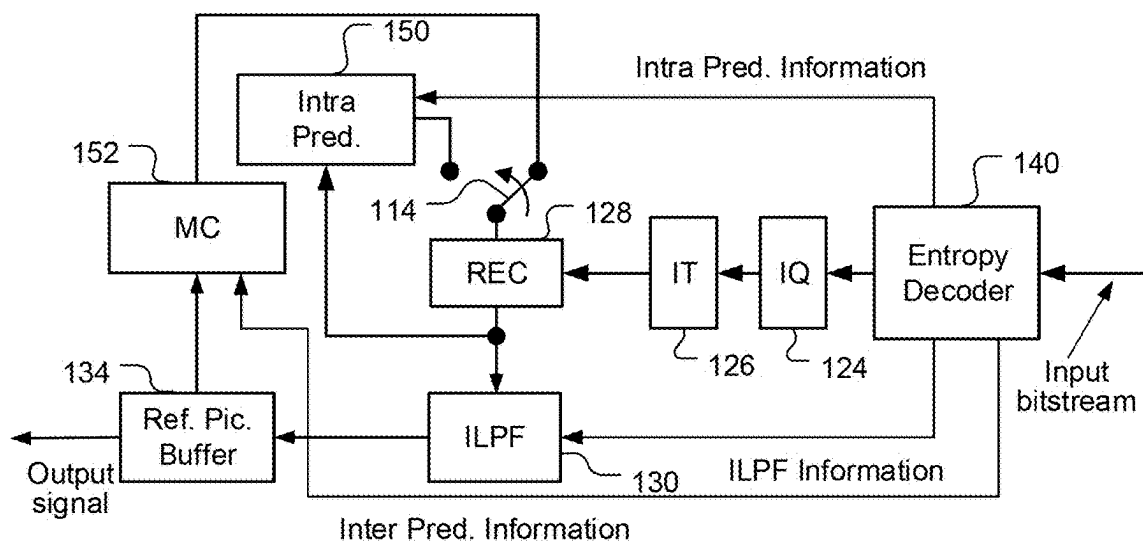


Fig. 1B

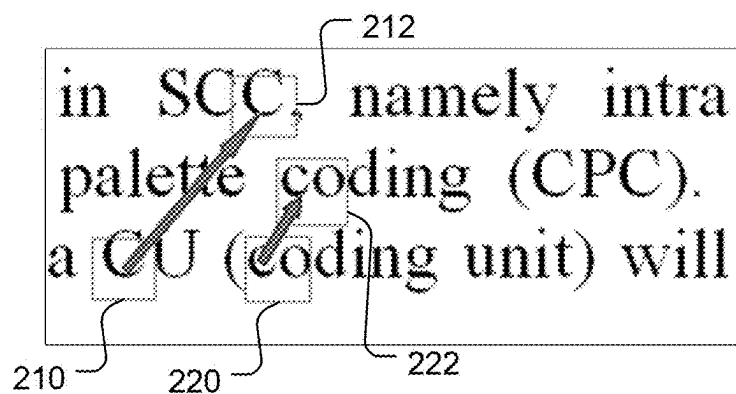


Fig. 2

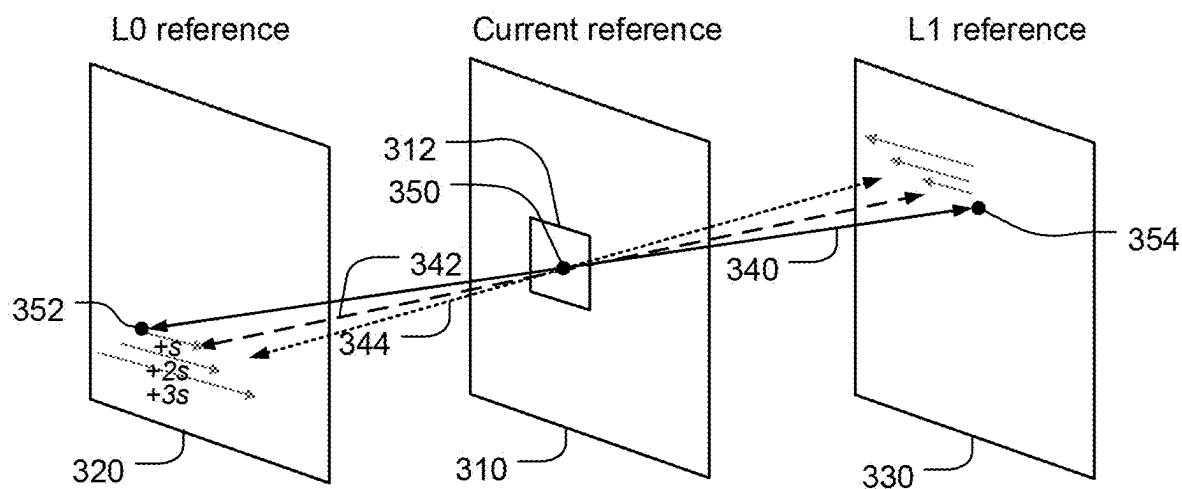


Fig. 3

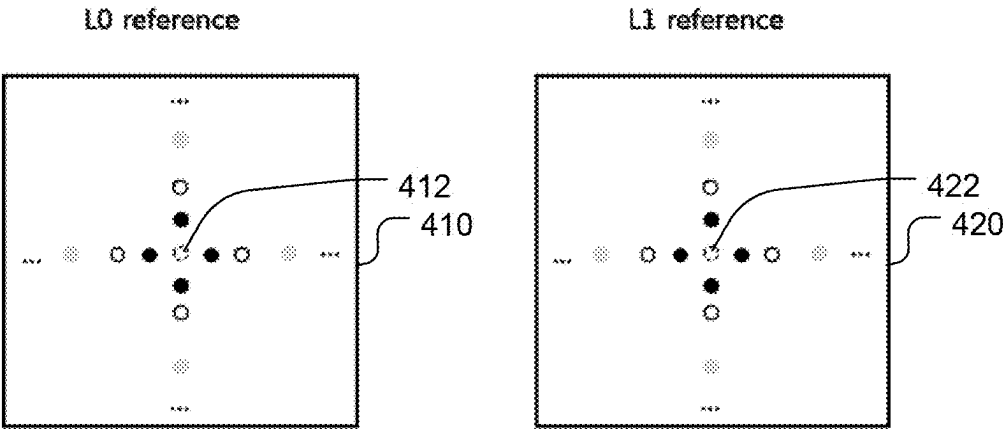


Fig. 4

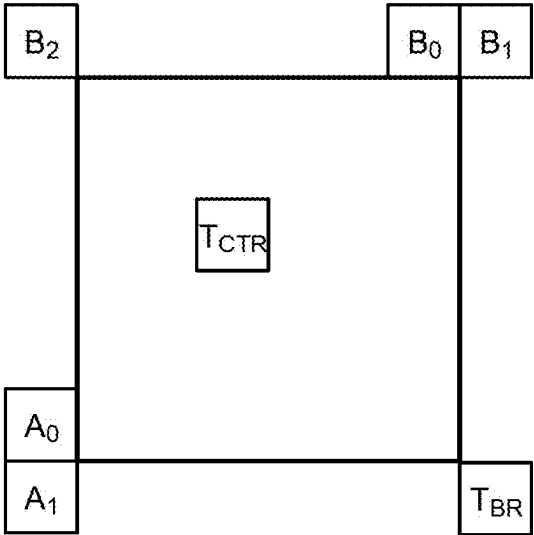


Fig. 5

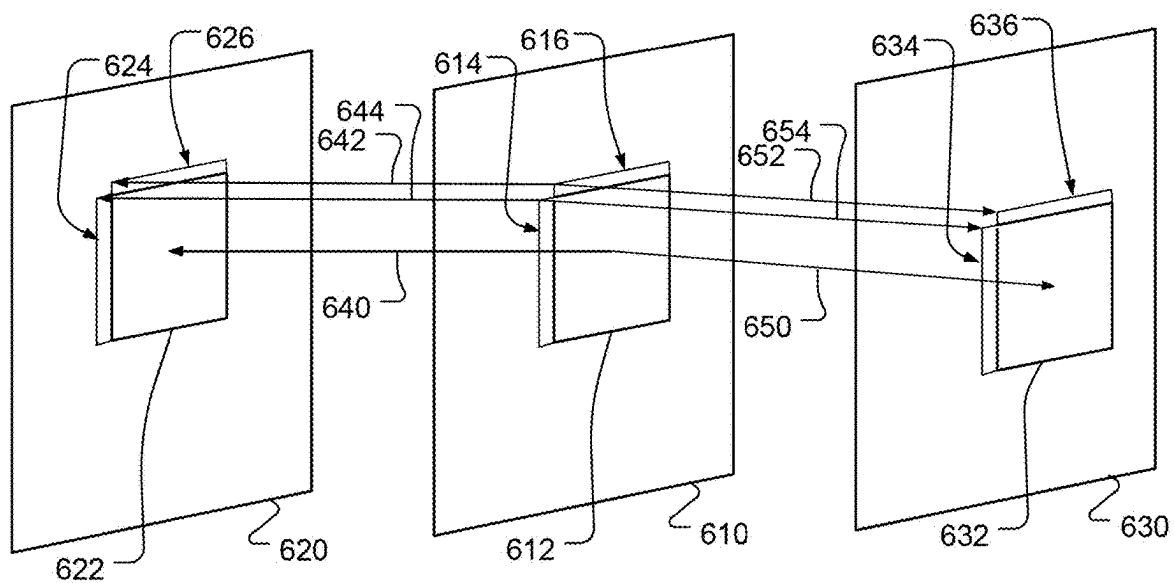


Fig. 6

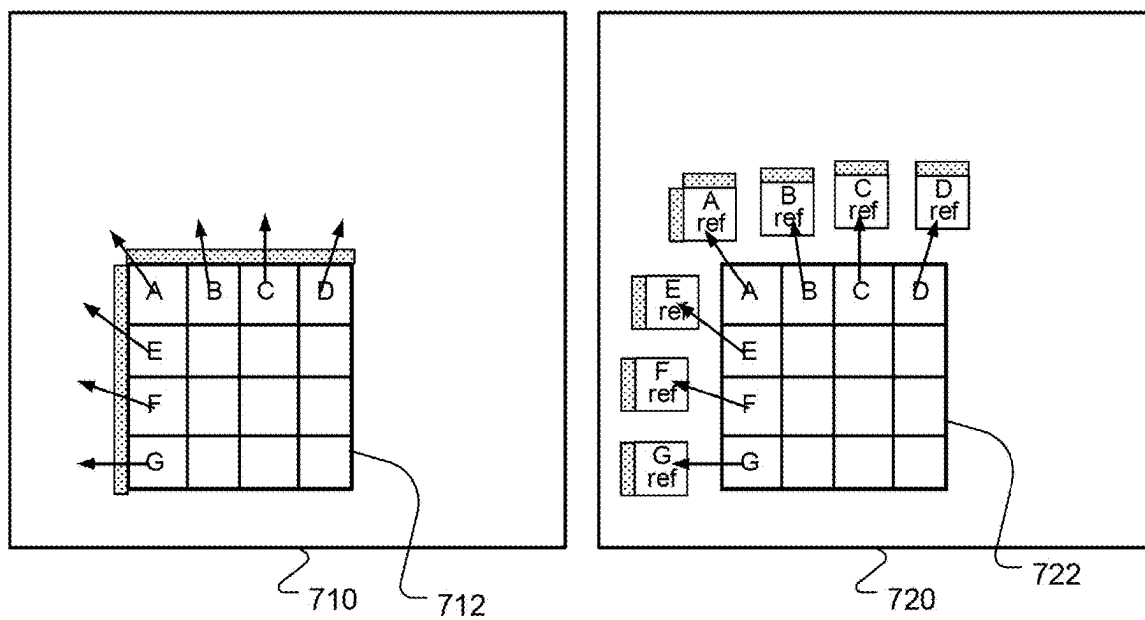
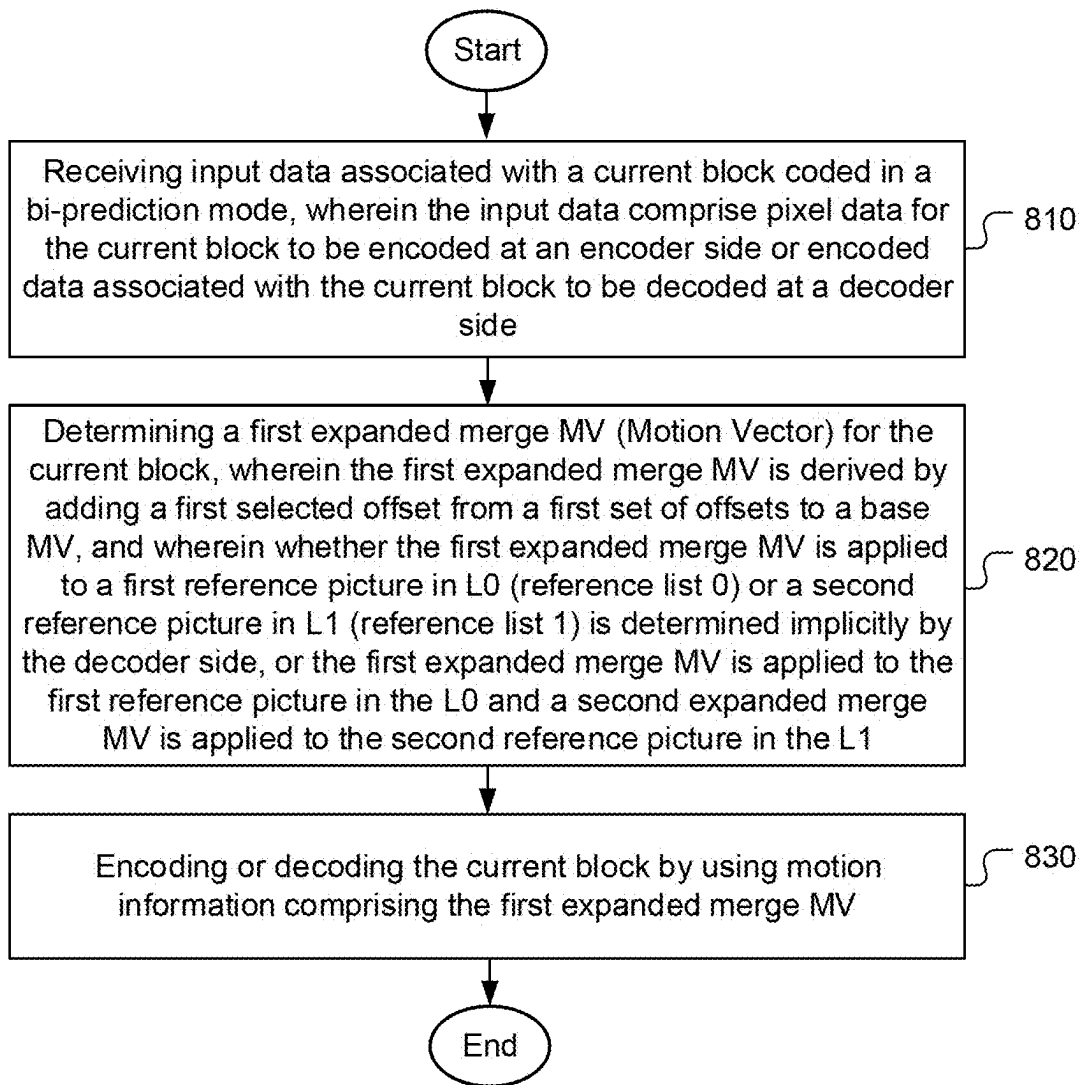


Fig. 7

**Fig. 8**

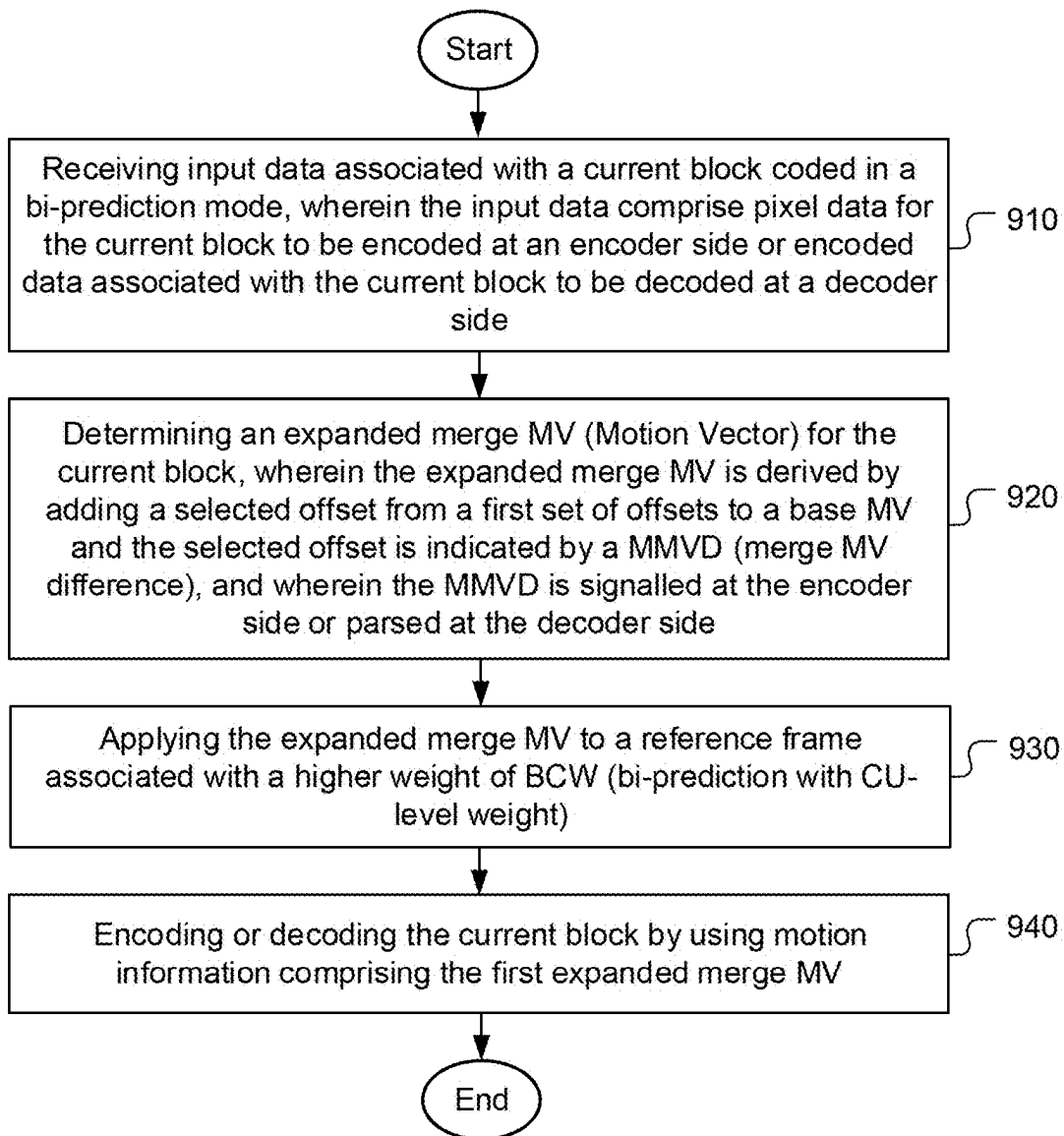


Fig. 9

METHOD AND APPARATUS FOR IMPROVEMENT OF VIDEO CODING USING MERGE WITH MVD MODE WITH TEMPLATE MATCHING

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present invention is a non-Provisional application of and claims priority to U.S. Provisional Patent Application No. 63/336,389, filed on Apr. 29, 2022. The U.S. Provisional patent application is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to video coding system using MMVD (Merge mode Motion Vector Difference) coding tool. In particular, the present invention relates to adding flexibility to MMVD design so as to improve coding performance.

BACKGROUND

[0003] Versatile video coding (VVC) is the latest international video coding standard developed by the Joint Video Experts Team (JVET) of the ITU-T Video Coding Experts Group (VCEG) and the ISO/IEC Moving Picture Experts Group (MPEG). The standard has been published as an ISO standard: ISO/IEC 23090-3:2021, Information technology-Coded representation of immersive media-Part 3: Versatile video coding, published February 2021. VVC is developed based on its predecessor HEVC (High Efficiency Video Coding) by adding more coding tools to improve coding efficiency and also to handle various types of video sources including 3-dimensional (3D) video signals.

[0004] FIG. 1A illustrates an exemplary adaptive Inter/Intra video coding system incorporating loop processing. For Intra Prediction, the prediction data is derived based on previously coded video data in the current picture. For Inter Prediction 112, Motion Estimation (ME) is performed at the encoder side and Motion Compensation (MC) is performed based on the result of ME to provide prediction data derived from other picture(s) and motion data. Switch 114 selects Intra Prediction 110 or Inter-Prediction 112 and the selected prediction data is supplied to Adder 116 to form prediction errors, also called residues. The prediction error is then processed by Transform (T) 118 followed by Quantization (Q) 120. The transformed and quantized residues are then coded by Entropy Encoder 122 to be included in a video bitstream corresponding to the compressed video data. The bitstream associated with the transform coefficients is then packed with side information such as motion and coding modes associated with Intra prediction and Inter prediction, and other information such as parameters associated with loop filters applied to underlying image area. The side information associated with Intra Prediction 110, Inter prediction 112 and in-loop filter 130, are provided to Entropy Encoder 122 as shown in FIG. 1A. When an Inter-prediction mode is used, a reference picture or pictures have to be reconstructed at the encoder end as well. Consequently, the transformed and quantized residues are processed by Inverse Quantization (IQ) 124 and Inverse Transformation (IT) 126 to recover the residues. The residues are then added back to prediction data 136 at Reconstruction (REC) 128 to recon-

struct video data. The reconstructed video data may be stored in Reference Picture Buffer 134 and used for prediction of other frames.

[0005] As shown in FIG. 1A, incoming video data undergoes a series of processing in the encoding system. The reconstructed video data from REC 128 may be subject to various impairments due to a series of processing. Accordingly, in-loop filter 130 is often applied to the reconstructed video data before the reconstructed video data are stored in the Reference Picture Buffer 134 in order to improve video quality. For example, deblocking filter (DF), Sample Adaptive Offset (SAO) and Adaptive Loop Filter (ALF) may be used. The loop filter information may need to be incorporated in the bitstream so that a decoder can properly recover the required information. Therefore, loop filter information is also provided to Entropy Encoder 122 for incorporation into the bitstream. In FIG. 1A, Loop filter 130 is applied to the reconstructed video before the reconstructed samples are stored in the reference picture buffer 134. The system in FIG. 1A is intended to illustrate an exemplary structure of a typical video encoder. It may correspond to the High Efficiency Video Coding (HEVC) system, VP8, VP9, H.264 or VVC.

[0006] The decoder, as shown in FIG. 1B, can use similar or portion of the same functional blocks as the encoder except for Transform 118 and Quantization 120 since the decoder only needs Inverse Quantization 124 and Inverse Transform 126. Instead of Entropy Encoder 122, the decoder uses an Entropy Decoder 140 to decode the video bitstream into quantized transform coefficients and needed coding information (e.g. ILPF information, Intra prediction information and Inter prediction information). The Intra prediction 150 at the decoder side does not need to perform the mode search. Instead, the decoder only needs to generate Intra prediction according to Intra prediction information received from the Entropy Decoder 140. Furthermore, for Inter prediction, the decoder only needs to perform motion compensation (MC 152) according to Inter prediction information received from the Entropy Decoder 140 without the need for motion estimation.

[0007] According to VVC, an input picture is partitioned into non-overlapped square block regions referred as CTUs (Coding Tree Units), similar to HEVC. Each CTU can be partitioned into one or multiple smaller size coding units (CUs). The resulting CU partitions can be in square or rectangular shapes. Also, VVC divides a CTU into prediction units (PUS) as a unit to apply prediction process, such as Inter prediction, Intra prediction, etc.

[0008] The VVC standard incorporates various new coding tools to further improve the coding efficiency over the HEVC standard. Among various new coding tools, some coding tools relevant to the present invention are reviewed as follows. For example, Merge with MVD Mode (MMVD) technique re-uses the same merge candidates as those in VVC and a selected candidate can be further expanded by a motion vector expression method. It is desirable to develop techniques to reduce the complexity of MMVD.

BRIEF SUMMARY OF THE INVENTION

[0009] A method and apparatus for video coding using MMVD (Merge with MVD (Motion Vector Difference)) mode are disclosed. According to the method, input data associated with a current block coded in a bi-prediction mode are received, where the input data comprise pixel data

for the current block to be encoded at an encoder side or encoded data associated with the current block to be decoded at a decoder side. A first expanded merge MV (Motion Vector) for the current block is determined where the first expanded merge MV is derived by adding a first selected offset from a first set of offsets to a base MV. Whether the first expanded merge MV is applied to a first reference picture in L0 (reference list 0) or a second reference picture in L1 (reference list 1) is determined implicitly by the decoder side, or the first expanded merge MV is applied to the first reference picture in the L0 and a second expanded merge MV is applied to the second reference picture in the L1. The current block is encoded or decoded by using motion information comprising the first expanded merge MV.

[0010] In one embodiment, whether the first expanded merge MV is applied to the first reference picture in the L0 or the L1 is determined according to a matching cost measured between one or more first neighbouring areas of the current block and one or more second neighbouring areas of a first reference block in the L0 or the L1. Said one or more first neighbouring areas of the current block comprise a first top neighbouring area and a first left neighbouring area of the current block and said one or more second neighbouring areas of the first reference block comprise a second top neighbouring area and a second left neighbouring area of the first reference block. The matching cost is only calculated for the first reference picture in the L0 (L1) and is disregarded for the first reference picture in the L1 (L0) if the first expanded merge MV is applied to the first reference picture in the L0 (L1).

[0011] In one embodiment, one or more syntaxes related to a MVD (MV difference) between the first expanded merge MV and the based MV is signalled at the encoder side or parsed at the decoder side. When the first expanded merge MV is applied to the first reference picture in one of L0 and L1, the second reference picture in the other of the L0 and the L1 uses a scaled MVD or a clipped and scaled MVD signalled at the encoder side or parsed at the decoder side.

[0012] In one embodiment, the second expanded merge MV is derived by adding a second selected offset from a second set of offsets to the base MV. In one embodiment, M first expanded merge MV candidates corresponding to a portion of a set of first expanded merge MV candidates are selected and N second expanded merge MV candidates corresponding to a portion of a set of second expanded merge MV candidates are selected according to matching costs associated with the set of first expanded merge MV candidates and the set of second expanded merge MV candidates, and wherein M and N are positive integers. $M \times N$ joint expanded merge MV candidates can be generated from the M first expanded merge MV candidates and the N second expanded merge MV candidates. The $M \times N$ joint expanded merge MV candidates are then reordered according to the matching costs. The first expanded merge MV and the second expanded merge MV can be selected from K best joint expanded merge MV candidates among the $M \times N$ joint expanded merge MV candidates according to the matching costs, and K is smaller than $M \times N$. In one embodiment, M and N correspond to predetermined numbers, adaptively varying numbers based on matching cost distribution, adaptively varying numbers based on BCW (Bi-prediction with CU-level Weights) index, or explicitly signalled values.

[0013] According to another method, an expanded merge MV (Motion Vector) for the current block is determined by adding a selected offset from a first set of offsets to a base MV and the selected offset is indicated by a MMVD (merge MV difference), and the MMVD is signalled at the encoder side or parsed at the decoder side. The expanded merge MV is always applied to a reference frame associated with a higher weight of BCW (bi-prediction with CU-level weight).

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1A illustrates an exemplary adaptive Inter/Intra video coding system incorporating loop processing.

[0015] FIG. 1B illustrates a corresponding decoder for the encoder in FIG. 1A.

[0016] FIG. 2 illustrates an example of CPR (Current Picture Referencing) compensation, where blocks are predicted by corresponding blocks in the same picture.

[0017] FIG. 3 illustrates an example of MMVD (Merge mode Motion Vector Difference) search process, where a current block in the current frame is processed by bi-direction prediction using a L0 reference frame and a L1 reference frame.

[0018] FIG. 4 illustrates the offset distances in the horizontal and vertical directions for a L0 reference block and L1 reference block according to MMVD.

[0019] FIG. 5 illustrates an example of merge mode candidate derivation from spatial and temporal neighbouring blocks.

[0020] FIG. 6 illustrates an example of templates used for the current block and corresponding reference blocks to measure matching costs associated with merge candidates.

[0021] FIG. 7 illustrates an example of template and reference samples of the template for block with sub-block motion using the motion information of the subblocks of the current block.

[0022] FIG. 8 illustrates a flowchart of an exemplary video coding system that utilizes flexible MMVD design to improve the coding performance according to an embodiment of the present invention.

[0023] FIG. 9 illustrates a flowchart of another exemplary video coding system that utilizes separate MVDs for reference pictures in different reference lists according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0024] It will be readily understood that the components of the present invention, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the systems and methods of the present invention, as represented in the figures, is not intended to limit the scope of the invention, as claimed, but is merely representative of selected embodiments of the invention. References throughout this specification to “one embodiment,” “an embodiment,” or similar language mean that a particular feature, structure, or characteristic described in connection with the embodiment may be included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodi-

ment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment.

[0025] Furthermore, the described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. One skilled in the relevant art will recognize, however, that the invention can be practiced without one or more of the specific details, or with other methods, components, etc. In other instances, well-known structures, or operations are not shown or described in detail to avoid obscuring aspects of the invention. The illustrated embodiments of the invention will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. The following description is intended only by way of example, and simply illustrates certain selected embodiments of apparatus and methods that are consistent with the invention as claimed herein.

Current Picture Referencing

[0026] Motion Compensation, one of the key technologies in hybrid video coding, explores the pixel correlation between adjacent pictures. It is generally assumed that, in a video sequence, the patterns corresponding to objects or background in a frame are displaced to form corresponding objects in the subsequent frame or correlated with other patterns within the current frame. With the estimation of such displacement (e.g. using block matching techniques), the pattern can be mostly reproduced without the need to re-code the pattern. Similarly, block matching and copy has also been tried to allow selecting the reference block from the same picture as the current block. It was observed to be inefficient when applying this concept to camera captured videos. Part of the reasons is that the textual pattern in a spatial neighbouring area may be similar to the current coding block, but usually with some gradual changes over the space. It is difficult for a block to find an exact match within the same picture in a video captured by a camera. Accordingly, the improvement in coding performance is limited.

[0027] However, the situation for spatial correlation among pixels within the same picture is different for screen contents. For a typical video with texts and graphics, there are usually repetitive patterns within the same picture. Hence, intra (picture) block compensation has been observed to be very effective. A new prediction mode, i.e., the intra block copy (IBC) mode or called current picture referencing (CPR), has been introduced for screen content coding to utilize this characteristic. In the CPR mode, a prediction unit (PU) is predicted from a previously reconstructed block within the same picture. Further, a displacement vector (called block vector or BV) is used to indicate the relative displacement from the position of the current block to that of the reference block. The prediction errors are then coded using transformation, quantization and entropy coding. An example of CPR compensation is illustrated in FIG. 2, where block 212 is a corresponding block for block 210, and block 222 is a corresponding block for block 220. In this technique, the reference samples correspond to the reconstructed samples of the current decoded picture prior to in-loop filter operations, both deblocking and sample adaptive offset (SAO) filters in HEVC.

[0028] The very first version of CPR was proposed in JCTVC-M0350 (Budagavi et al., *AHG8: Video coding using Intra motion compensation*, Joint Collaborative Team on

Video Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC 1/SC 29/WG11, 13th Meeting: Incheon, KR, 18-26 Apr. 2013, Document: JCTVC-M0350) to the HEVC Range Extensions (RExt) development. In this version, the CPR compensation was limited to be within a small local area, with only 1-D block vector and only for block size of $2N \times 2N$. Later, a more advanced CPR design has been developed during the standardization of HEVC SCC (Screen Content Coding).

[0029] When CPR is used, only part of the current picture can be used as the reference picture. A few bitstream conformance constraints are imposed to regulate the valid MV value referring to the current picture. First, one of the following two must be true:

$$BV_x + offsetX + nPbSw + xPbs - xCbs \leq 0 \quad (1)$$

$$BV_y + offsetY + nPbSh + yPbs - yCbs \leq 0 \quad (2)$$

[0030] Second, the following WPP condition must be true:

$$(xCbs + BV_x + offsetX + nPbSw - 1) / CtbSizeY - xCbs / CtbSizeY \leq \quad (3)$$

$$yCbs / CtbSizeY - (yPbs + BV_y + offsetY + nPbSh - 1) / CtbSizeY$$

[0031] In equations (1) through (3), (BV_x, BV_y) is the luma block vector (the motion vector for CPR) for the current PU; $nPbSw$ and $nPbSh$ are the width and height of the current PU; $(xPbs, yPbs)$ is the location of the top-left pixel of the current PU relative to the current picture; $(xCbs, yCbs)$ is the location of the top-left pixel of the current CU relative to the current picture; and $CtbSizeY$ is the size of the CTU. $offsetX$ and $offsetY$ are two adjusted offsets in two dimensions in consideration of chroma sample interpolation for the CPR mode.

$$offsetX = BVC_x \& 0 \times 7 \& 2:0 \quad (4)$$

$$offsetY = BVC_y \& 0 \times 7 \& 2:0 \quad (5)$$

[0032] (BVC_x, BVC_y) is the chroma block vector, in 1/8-pel resolution in HEVC.

[0033] Third, the reference block for CPR must be within the same tile/slice boundary.

Merge with MVD Mode (MMVD) Technique

[0034] The MMVD technique is proposed in JVECT-J0024. MMVD is used for either skip or merge modes with a proposed motion vector expression method. MMVD re-uses the same merge candidates as those in VVC. Among the merge candidates, a candidate can be selected, and is further expanded by the proposed motion vector expression method. MMVD provides a new motion vector expression with simplified signalling. The expression method includes prediction direction information, starting point (also referred as a base in this disclosure), motion magnitude (also referred as a distance in this disclosure), and motion direction. FIG. 3 illustrates an example of MMVD search process, where a current block 312 in the current frame 310 is processed by bi-direction prediction using a L0 reference frame 320 and a L1 reference frame 330. A pixel location 350 is projected

to pixel location **352** in L0 reference frame **320** and pixel location **354** in L1 reference frame **330**. According to the MMVD search process, updated locations will be searched by adding offsets in selected directions. For example, the updated locations correspond to locations along line **342** or **344** in the horizontal direction with distances to at s, 2 s or 3 s.

[0035] This proposed technique uses a merge candidate list as is. However, only candidates which are default merge type (i.e., MRG_TYPE_DEFAULT_N) are considered for MMVD's expansion. Prediction direction information indicates a prediction direction among L0, L1, and L0 and L1 predictions. In B slice, the proposed method can generate bi-prediction candidates from merge candidates with uni-prediction by using mirroring technique. For example, if a merge candidate is uni-prediction with L1, a reference index of L0 is decided by searching a reference picture in list 0, which is mirrored with the reference picture for list 1. If there is no corresponding picture, the nearest reference

TABLE 1

Base candidate IDX				
Base candidate IDX	0	1	2	3
Nth MVP	1st MVP	2nd MVP	3rd MVP	4th MVP

[0039] Distance index specifies motion magnitude information and indicates the pre-defined offset from the starting points (**412** and **422**) for a L0 reference block **410** and L1 reference block **420** as shown in FIG. 4. In FIG. 4, an offset is added to either the horizontal component or the vertical component of the starting MV, where small circles in different styles correspond to different offsets from the centre. The relation between the distance index and pre-defined offset is specified in Table 2.

TABLE 2

Distance IDX								
Distance IDX	0	1	2	3	4	5	6	7
Pixel distance	1/4-pel	1/2-pel	1-pel	2-pel	4-pel	8-pel	16-pel	32-pel

picture to the current picture is used. L0' MV is derived by scaling L1's MV and the scaling factor is calculated by POC distance.

[0036] In MMVD, after a merge candidate is selected, it is further expanded or refined by the signalled MVDs information. The further information includes a merge candidate flag, an index to specify motion magnitude, and an index for indication of the motion direction. In MMVD mode, one of the first two candidates in the merge list is selected to be used as an MV basis. The MMVD candidate flag is signalled to specify which one is used between the first and second merge candidates. The initial MVs (i.e., merge candidates) selected from the merge candidate list are also referred as bases in this disclosure. After searching the set of locations, a selected MV candidate is referred as an expanded MV candidate in this disclosure.

[0037] If the prediction direction of the MMVD candidate is the same as one of the original merge candidate, the index with value 0 is signalled as the MMVD prediction direction. Otherwise, the index with value 1 is signalled. After sending first bit, the remaining prediction direction is signalled based on the pre-defined priority order of MMVD prediction direction. Priority order is L0/L1 prediction, L0 prediction and L1 prediction. If the prediction direction of merge candidate is L1, signalling '0' indicates MMVD' prediction direction as L1. Signalling '10' indicates MMVD' prediction direction as L0 and L1. Signalling '11' indicates MMVD' prediction direction as L0. If L0 and L1 prediction lists are same, MMVD's prediction direction information is not signalled.

[0038] Base candidate index, as shown in Table 1, defines the starting point. Base candidate index indicates the best candidate among candidates in the list as follows.

[0040] Direction index represents the direction of the MVD relative to the starting point. The direction index can represent of the four directions as shown below. Direction index represents the direction of the MVD relative to the starting point. The direction index can represent the four directions as shown in Table 3. It is noted that the meaning of MVD sign could be variant according to the information of starting MVs. When the starting MVs are an un-prediction MV or bi-prediction MVs with both lists pointing to the same side of the current picture (i.e. POCs of two references both larger than the POC of the current picture, or both smaller than the POC of the current picture), the sign in Table 3 specifies the sign of the MV offset added to the starting MV. When the starting MVs are bi-prediction MVs with the two MVs pointing to the different sides of the current picture (i.e. the POC of one reference larger than the POC of the current picture, and the POC of the other reference smaller than the POC of the current picture), and the difference of POC in list 0 is greater than the one in list 1, the sign in Table 3 specifies the sign of MV offset added to the list0 MV component of the starting MV and the sign for the list1 MV has an opposite value. Otherwise, if the difference of POC in list 1 is greater than list 0, the sign in Table 3 specifies the sign of the MV offset added to the list1 MV component of starting MV and the sign for the list0 MV has an opposite value.

TABLE 3

Direction IDX				
Direction IDX	00	01	10	11
x-axis	+	-	N/A	N/A
y-axis	N/A	N/A	+	-

[0041] To reduce the encoder complexity, block restriction is applied. If either width or height of a CU is less than 4, MMVD is not performed.

Multi-Hypothesis Prediction (MH) Technique

[0042] Multi-hypothesis prediction is proposed to improve the existing prediction modes in inter pictures, including uni-prediction of advanced motion vector prediction (AMVP) mode, skip and merge mode, and intra mode. The general concept is to combine an existing prediction mode with an extra merge indexed prediction. The merge indexed prediction is performed in a manner the same as that for the regular merge mode, where a merge index is signalled to acquire motion information for the motion compensated prediction. The final prediction is the weighted average of the merge indexed prediction and the prediction generated by the existing prediction mode, where different weights are applied depending on the combinations. Detail information can be found in JVET-K1030 (Chih-Wei Hsu, et al., Description of Core Experiment 10: Combined and multi-hypothesis prediction, Joint Collaborative Team on Video Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC 1/SC 29/WG11, 11th Meeting: Ljubljana, SI, 10-18 Jul. 2018, Document: JVET-K1030), or JVET-L0100 (Man-Shu Chiang, et al., CE10.1.1: Multi-hypothesis prediction for improving AMVP mode, skip or merge mode, and intra mode, Joint Collaborative Team on Video Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC 1/SC 29/WG11, 12th Meeting: Macao, CN, 3-12 Oct. 2018, Document: JVET-L0100).

Pairwise Averaged Merge Candidates

[0043] Pairwise average candidates are generated by averaging predefined pairs of candidates in the current merge candidate list, and the predefined pairs are defined as $\{(0, 1), (0, 2), (1, 2), (0, 3), (1, 3), (2, 3)\}$, where the numbers denote the merge indices to the merge candidate list. The averaged motion vectors are calculated separately for each reference list. If both motion vectors are available in one list, these two motion vectors are averaged even when they point to different reference pictures; if only one motion vector is available, use the one directly; if no motion vector is available, treat this list as invalid.

Merge Mode

[0044] To increase the coding efficiency of motion vector (MV) coding in HEVC, HEVC has the Skip, and Merge mode. Skip and Merge modes obtain the motion information from spatially neighbouring blocks (spatial candidates) or a temporal co-located block (temporal candidate). When a PU is Skip or Merge mode, no motion information is coded, instead, only the index of the selected candidate is coded. For Skip mode, the residual signal is forced to be zero and not coded. In HEVC, if a particular block is encoded as Skip or Merge, a candidate index is signalled to indicate which candidate among the candidate set is used for merging. Each merged PU reuses the MV, prediction direction, and reference picture index of the selected candidate.

[0045] For Merge mode in HM-4.0 in HEVC, as shown in FIG. 5, up to four spatial MV candidates are derived from A_0 , A_1 , B_0 and B_1 , and one temporal MV candidate is derived from TBR or TCTR (TBR is used first, if TBR is not available, TCTR is used instead). Note that if any of the four

spatial MV candidates is not available, the position B_2 is then used to derive another MV candidate as a replacement. After the derivation process of the four spatial MV candidates and one temporal MV candidate, removing redundancy (pruning) is applied to remove redundant MV candidates. If after removing redundancy (pruning), the number of available MV candidates is smaller than five, three types of additional candidates are derived and added to the candidate set (candidate list). The encoder selects one final candidate within the candidate set for Skip or Merge modes based on the rate-distortion optimization (RDO) decision, and transmits the index to the decoder.

[0046] Hereafter, we will denote the skip and merge mode as “merge mode”. In other words, when the “merge mode” is mentioned in the following specification, the “merge mode” may refer to both skip and merge modes.

Adaptive Reordering of Merge Candidates with Template Matching (ARMC-TM)

[0047] The merge candidates are adaptively reordered according to costs evaluated using template matching (TM). The reordering method can be applied to the regular merge mode, template matching (TM) merge mode, and affine merge mode (excluding the SbTMVP candidate). For the TM merge mode, merge candidates are reordered before the refinement process.

[0048] After a merge candidate list is constructed, merge candidates are divided into multiple subgroups. The subgroup size is set to 5 for the regular merge mode and TM merge mode. The subgroup size is set to 3 for the affine merge mode. Merge candidates in each subgroup are reordered ascendingly according to cost values based on template matching. For ARMC-TM, the candidates in a subgroup are skipped if the subgroup satisfies the following 2 conditions: (1) the subgroup is the last subgroup and (2) the subgroup is not the first subgroup. For simplification, merge candidates in the last but not the first subgroup are not reordered.

[0049] The template matching cost of a merge candidate is measured as the sum of absolute differences (SAD) between samples of a template of the current block and their corresponding reference samples. The template comprises a set of reconstructed samples neighbouring to the current block. Reference samples of the template are located by the motion information of the merge candidate.

[0050] When a merge candidate utilizes bi-directional prediction, the reference samples of the template of the merge candidate are also generated by bi-prediction as shown in FIG. 6. In FIG. 6, block 612 corresponds to a current block in current picture 610, blocks 622 and 632 correspond to reference blocks in reference pictures 620 and 630 in list 0 and list 1 respectively. Templates 614 and 616 are for current block 612, templates 624 and 626 are for reference block 622, and templates 634 and 636 are for reference block 632. Motion vectors 640, 642 and 644 are merge candidates in list 0 and motion vectors 650, 652 and 654 are merge candidates in list 1.

[0051] For subblock-based merge candidates with subblock size equal to $W_{sub} \times H_{sub}$, the above template comprises several sub-templates with the size of $W_{sub} \times 1$, and the left template comprises several sub-templates with the size of $1 \times H_{sub}$. As shown in FIG. 7, the motion information of the subblocks in the first row and the first column of current block is used to derive the reference samples of each sub-template. In FIG. 7, block 712 corresponds to a current

block in current picture **710** and block **722** corresponds to a collocated block in reference picture **720**. Each small square in the current block and the collocated block corresponds to a subblock. The dot-filled areas on the left and top of the current block correspond to template for the current block. The boundary subblocks are labelled from A to G. The arrow associated with each subblock corresponds to the motion vector of the subblock. The reference subblocks (labelled as Aref to Gref) are located according to the motion vectors associated with the boundary subblocks.

Improvement of MMVD with Template Matching

[0052] In some designs of MMVD with template matching (TM), for each base MV, it selects K MVD candidates from a total number of S*D combinations of S steps and D directions. The selection is based on the TM cost. If bi-directional prediction is used, the signalled MVD is implicitly applied to the reference frame with a larger temporal distance. For the other reference frame, the applied MVD is the signalled MVD, but down-scaled according to the temporal distance difference. In such design, the MVD selection for the bi-directional case is lack of freedom. The following proposed methods will improve MMVD in this aspect.

[0053] In one method, if bi-directional prediction is used in MMVD, the signalled MVD is implicitly applied to the reference frame with a higher weight of bi-prediction with CU-level weight (BCW).

[0054] In another method, if bi-directional prediction is used in MMVD, the reference frame that the signalled MVD is applied to is determined by the TM cost. Specifically, two TM costs are derived by applying the signalled MVD to one reference frame at a time, and the signalled MVD is finally applied to the reference frame with the lower TM cost. In one embodiment, when two TM costs are derived by applying the signalled MVD to one reference frame at a time, the other frame applies the scaled signalled MVD. In another embodiment, the other frame applies the clipped scaled signalled MVD. In another embodiment, the other frame is not considered in TM cost computation.

[0055] In another method, if bi-directional prediction is used in MMVD, two reference frames can have independent MVDs. Specifically, TM-based re-ordering is performed for each reference frame, and M candidates of the one reference frame and N candidates of the other reference frame are selected to form M*N bi-prediction candidates. Another TM-based re-ordering is performed on these M*N candidates to select K candidates for further signalling. Note that this method can be achieved without codeword change. The values of M and N can be predetermined fixed numbers, adaptively changed numbers based on the TM cost distribution, adaptively changed numbers based on BCW index, or explicitly signalled values.

[0056] In another method, if the base is bi-prediction, multiple uni-prediction candidates are generated by using only one of the reference frames. TM costs of all bi-prediction and uni-prediction candidates are computed and compared, where bi-prediction candidates can be generated by using the original MMVD design (i.e., S*D candidates) or the foregoing proposed design (i.e., M*N candidates), and uni-prediction candidates can be generated by considering all S*D candidates or just a subset of S*D candidates. After TM cost comparison, K candidates are selected from all candidates for further signalling according to one embodiment of the present invention. Note that this method can be achieved without codeword change.

[0057] In another method, if the base MV is uni-prediction, multiple bi-prediction candidates are generated by using the same reference frame with two different MVDs. TM costs of all uni-prediction and bi-prediction candidates are computed and compared, where uni-prediction candidates can be generated by considering all possible S*D candidates or just a subset of S*D candidates, and bi-prediction candidates can be generated by combining any two distinct uni-prediction candidates in S*D candidates, combining any two distinct uni-prediction candidates in a subset of S*D candidates, or combining two distinct uni-prediction candidates with one from a subset of S*D candidates and the other from another subset of S*D candidates. After TM cost comparison, select K candidates from all candidates for further signalling. Note that this method can be achieved without codeword change. Moreover, this method can be combined with the previous proposed method that generates uni-prediction candidates for bi-prediction bases.

[0058] Any of the MMVD methods described above can be implemented in encoders and/or decoders. For example, any of the proposed methods can be implemented in an inter coding module of an encoder (e.g. Inter Pred. **112** in FIG. **1A**), a motion compensation module (e.g., MC **152** in FIG. **1B**), a merge candidate derivation module of a decoder. Alternatively, any of the proposed methods can be implemented as a circuit coupled to the inter coding module of an encoder and/or motion compensation module, a merge candidate derivation module of the decoder. While the Inter-Pred. **112** and MC **152** are shown as individual processing units to support the MMVD methods, they may correspond to executable software or firmware codes stored on a media, such as hard disk or flash memory, for a CPU (Central Processing Unit) or programmable devices (e.g. DSP (Digital Signal Processor) or FPGA (Field Programmable Gate Array)).

[0059] FIG. **8** illustrates a flowchart of an exemplary video coding system that utilizes flexible MMVD design to improve the coding performance according to an embodiment of the present invention. The steps shown in the flowchart may be implemented as program codes executable on one or more processors (e.g., one or more CPUs) at the encoder side. The steps shown in the flowchart may also be implemented based hardware such as one or more electronic devices or processors arranged to perform the steps in the flowchart. According to this method, input data associated with a current block coded in a bi-prediction mode are received in step **810**, wherein the input data comprise pixel data for the current block to be encoded at an encoder side or prediction residual data associated with the current block to be decoded at a decoder side. A first expanded merge MV (Motion Vector) is determined for the current block in step **820**, wherein the first expanded merge MV is derived by adding a first selected offset from a first set of offsets to a base MV, and wherein whether the first expanded merge MV is applied to a first reference picture in L0 (reference list 0) or a second reference picture in L1 (reference list 1) is determined implicitly by the decoder side, or the first expanded merge MV is applied to the first reference picture in the L0 and a second expanded merge MV is applied to the second reference picture in the L1. The current block is encoded or decoded by using motion information comprising the first expanded merge MV in step **830**.

[0060] FIG. 9 illustrates a flowchart of another exemplary video coding system that utilizes separate MVDs for reference pictures in different reference lists according to an embodiment of the present invention. According to this method, input data associated with a current block coded in a bi-prediction mode are received in step 910, wherein the input data comprise pixel data for the current block to be encoded at an encoder side or prediction residual data associated with the current block to be decoded at a decoder side. An expanded merge MV (Motion Vector) is determined for the current block in step 920, wherein the expanded merge MV is derived by adding a selected offset from a first set of offsets to a base MV and the selected offset is indicated by a MMVD (merge MV difference), and wherein the MMVD is signalled at the encoder side or parsed at the decoder side. The expanded merge MV is applied to a reference frame associated with a higher weight of BCW (bi-prediction with CU-level weight) in step 930. The current block is encoded or decoded by using motion information comprising the first expanded merge MV in step 940.

[0061] The flowcharts shown are intended to illustrate an example of video coding according to the present invention. A person skilled in the art may modify each step, re-arranges the steps, split a step, or combine steps to practice the present invention without departing from the spirit of the present invention. In the disclosure, specific syntax and semantics have been used to illustrate examples to implement embodiments of the present invention. A skilled person may practice the present invention by substituting the syntax and semantics with equivalent syntax and semantics without departing from the spirit of the present invention.

[0062] The above description is presented to enable a person of ordinary skill in the art to practice the present invention as provided in the context of a particular application and its requirement. Various modifications to the described embodiments will be apparent to those with skill in the art, and the general principles defined herein may be applied to other embodiments. Therefore, the present invention is not intended to be limited to the particular embodiments shown and described, but is to be accorded the widest scope consistent with the principles and novel features herein disclosed. In the above detailed description, various specific details are illustrated in order to provide a thorough understanding of the present invention. Nevertheless, it will be understood by those skilled in the art that the present invention may be practiced.

[0063] Embodiment of the present invention as described above may be implemented in various hardware, software codes, or a combination of both. For example, an embodiment of the present invention can be one or more circuit circuits integrated into a video compression chip or program code integrated into video compression software to perform the processing described herein. An embodiment of the present invention may also be program code to be executed on a Digital Signal Processor (DSP) to perform the processing described herein. The invention may also involve a number of functions to be performed by a computer processor, a digital signal processor, a microprocessor, or field programmable gate array (FPGA). These processors can be configured to perform particular tasks according to the invention, by executing machine-readable software code or firmware code that defines the particular methods embodied by the invention. The software code or firmware code may be developed in different programming languages and dif-

ferent formats or styles. The software code may also be compiled for different target platforms. However, different code formats, styles and languages of software codes and other means of configuring code to perform the tasks in accordance with the invention will not depart from the spirit and scope of the invention.

[0064] The invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described examples are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

1. A method of video coding using MMVD (Merge with MVD (Motion Vector Difference)) mode, the method comprising:

receiving input data associated with a current block coded in a bi-prediction mode, wherein the input data comprise pixel data for the current block to be encoded at an encoder side or encoded data associated with the current block to be decoded at a decoder side;

determining a first expanded merge MV (Motion Vector) for the current block, wherein the first expanded merge MV is derived by adding a first selected offset from a first set of offsets to a base MV, and wherein whether the first expanded merge MV is applied to a first reference picture in L0 (reference list 0) or a second reference picture in L1 (reference list 1) is determined implicitly by the decoder side, or the first expanded merge MV is applied to the first reference picture in the L0 and a second expanded merge MV is applied to the second reference picture in the L1; and

encoding or decoding the current block by using motion information comprising the first expanded merge MV.

2. The method of claim 1, wherein whether the first expanded merge MV is applied to the first reference picture in the L0 or the L1 is determined according to a matching cost measured between one or more first neighbouring areas of the current block and one or more second neighbouring areas of a first reference block in the L0 or the L1.

3. The method of claim 2, wherein said one or more first neighbouring areas of the current block comprise a first top neighbouring area and a first left neighbouring area of the current block and said one or more second neighbouring areas of the first reference block comprise a second top neighbouring area and a second left neighbouring area of the first reference block.

4. The method of claim 2, wherein the matching cost is only calculated for the first reference picture in the L0 (L1) and is disregarded for the first reference picture in the L1 (L0) if the first expanded merge MV is applied to the first reference picture in the L0 (L1).

5. The method of claim 1, wherein one or more syntaxes related to a MVD (MV difference) between the first expanded merge MV and the based MV is signalled at the encoder side or parsed at the decoder side.

6. The method of claim 5, wherein when the first expanded merge MV is applied to the first reference picture in one of L0 and L1, the second reference picture in the other of the L0 and the L1 uses a scaled MVD signalled at the encoder side or parsed at the decoder side.

7. The method of claim 5, wherein when the first expanded merge MV is applied to the first reference picture

in one of L0 and L1, the first reference picture in the other of the L0 and the L1 uses a clipped and scaled MVD signalled at the encoder side or parsed at the decoder side.

8. The method of claim 1, wherein the second expanded merge MV is derived by adding a second selected offset from a second set of offsets to the base MV.

9. The method of claim 8, wherein M first expanded merge MV candidates corresponding to a portion of a set of first expanded merge MV candidates are selected and N second expanded merge MV candidates corresponding to a portion of a set of second expanded merge MV candidates are selected according to matching costs associated with the set of first expanded merge MV candidates and the set of second expanded merge MV candidates, and wherein M and N are positive integers.

10. The method of claim 9, wherein M×N joint expanded merge MV candidates are generated from the M first expanded merge MV candidates and the N second expanded merge MV candidates, and wherein the M×N joint expanded merge MV candidates are reordered according to the matching costs.

11. The method of claim 10, wherein the first expanded merge MV and the second expanded merge MV are selected from K best joint expanded merge MV candidates among the M×N joint expanded merge MV candidates according to the matching costs, and K is smaller than M×N.

12. The method of claim 11, wherein M and N correspond to predetermined numbers, adaptively varying numbers based on matching cost distribution, adaptively varying numbers based on BCW (Bi-prediction with CU-level Weights) index, or explicitly signalled values.

13. An apparatus for video coding using MMVD (Merge with MVD (Motion Vector Difference)) mode, the apparatus comprising one or more electronics or processors arranged to:

receive input data associated with a current block coded in a bi-prediction mode, wherein the input data comprise pixel data for the current block to be encoded at

an encoder side or encoded data associated with the current block to be decoded at a decoder side;

determine a first expanded merge MV (Motion Vector) for the current block, wherein the first expanded merge MV is derived by adding a first selected offset from a first set of offsets to a base MV, and wherein whether the first expanded merge MV is applied to a first reference picture in L0 (reference list 0) or a second reference picture in L1 (reference list 1) is determined implicitly by the decoder side, or the first expanded merge MV is applied to the first reference picture in the L0 and a second expanded merge MV is applied to the second reference picture in the L1; and

encode or decode the current block by using motion information comprising the first expanded merge MV.

14. A method of video coding using MMVD (Merge with MVD (Motion Vector Difference)) mode, the method comprising:

receiving input data associated with a current block coded in a bi-prediction mode, wherein the input data comprise pixel data for the current block to be encoded at an encoder side or encoded data associated with the current block to be decoded at a decoder side;

determining an expanded merge MV (Motion Vector) for the current block, wherein the expanded merge MV is derived by adding a selected offset from a first set of offsets to a base MV and the selected offset is indicated by a MMVD (merge MV difference), and wherein the MMVD is signalled at the encoder side or parsed at the decoder side; and

applying the expanded merge MV to a reference frame associated with a higher weight of BCW (bi-prediction with CU-level weight); and

encoding or decoding the current block by using motion information comprising the expanded merge MV.

15. (canceled)

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